

Cosmos

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What is COSMOS

- A testbed that allows researchers to explore real world scenarios
- Deployed in West Harlem, New York City
- Utilizes a multi-layer computing system which includes SDR nodes (user devices and radio hardware), edge cloud servers, and a general purpose cloud
- Currently supports 4 different experiments - plans to expand in the future

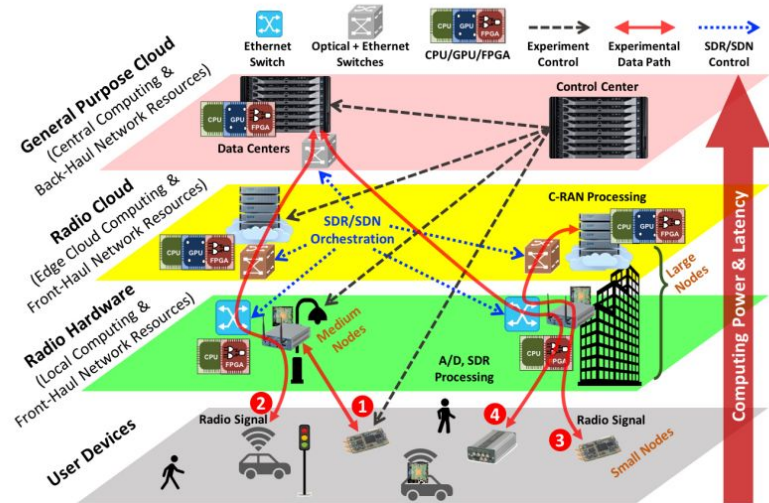
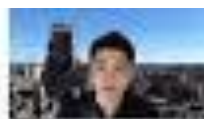


Figure 1: COSMOS' multi-layered computing architecture, where paths 1–4 demonstrate the data paths of example experiments with local/remote computing.

COSMOS: Envisioned Deployment



Large
(streetmap)



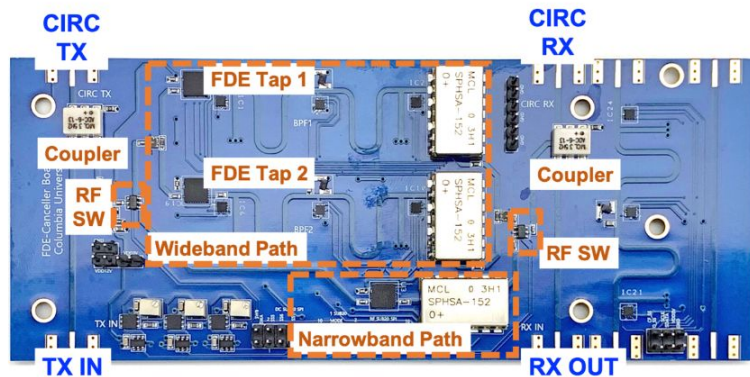
Medium
(street view)



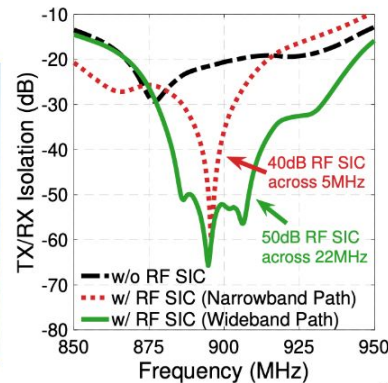
Small
(portable)

Experiment 1: Open-Access Full Duplex Wireless

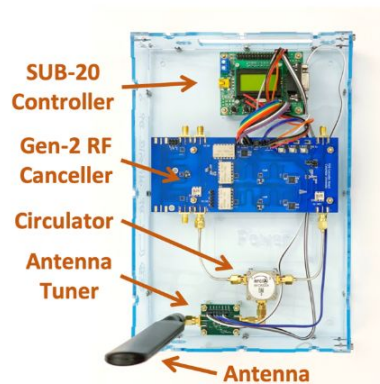
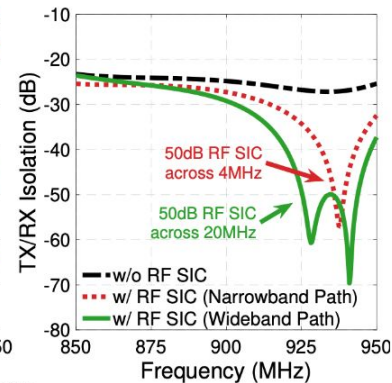
- Full duplex (FD) wireless allows simultaneous transmission and reception at the same frequency
- Able to double the network capacity at the physical layer
- Radio-frequency domain equalization used to improve performance and bandwidth of RF self-interference cancellation (SIC)
- Baseline programs include real time packet level digital SIC and packet decoding
- Measuring TX/RX isolation



(a)

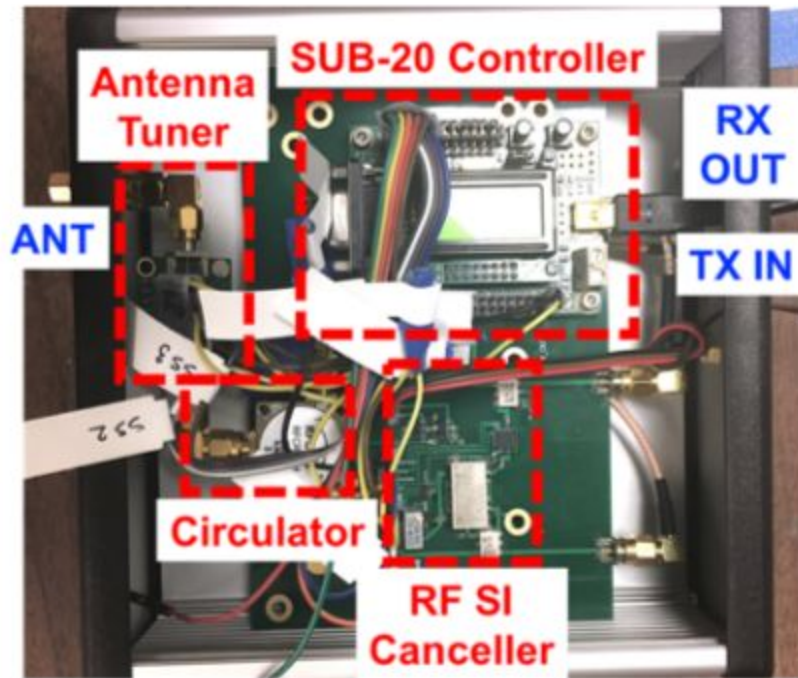


(b)



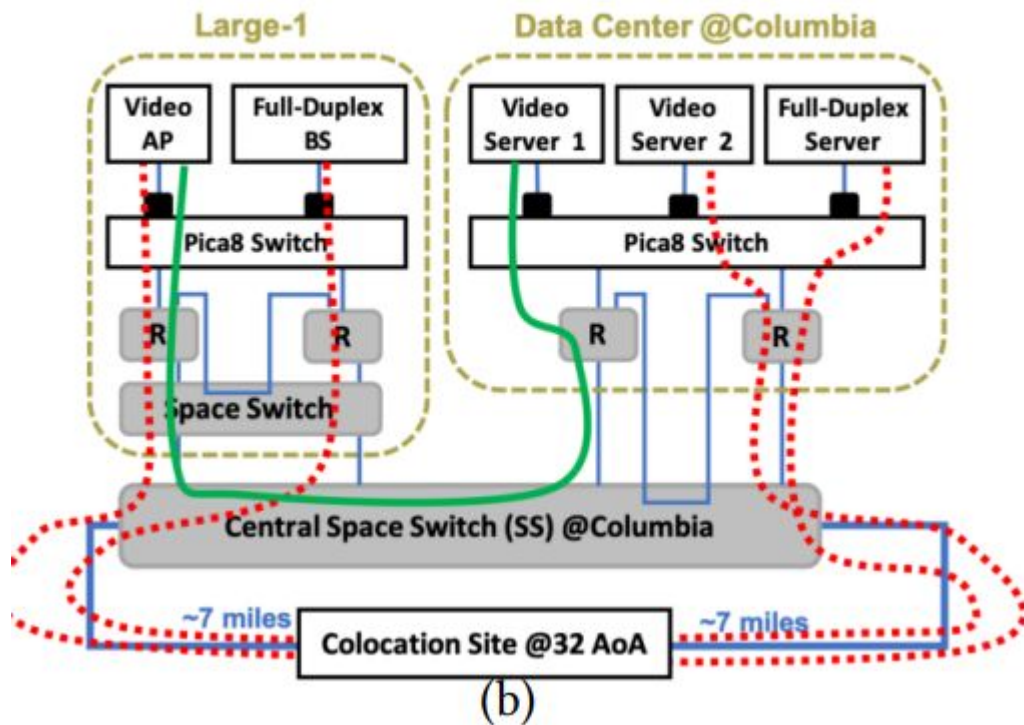
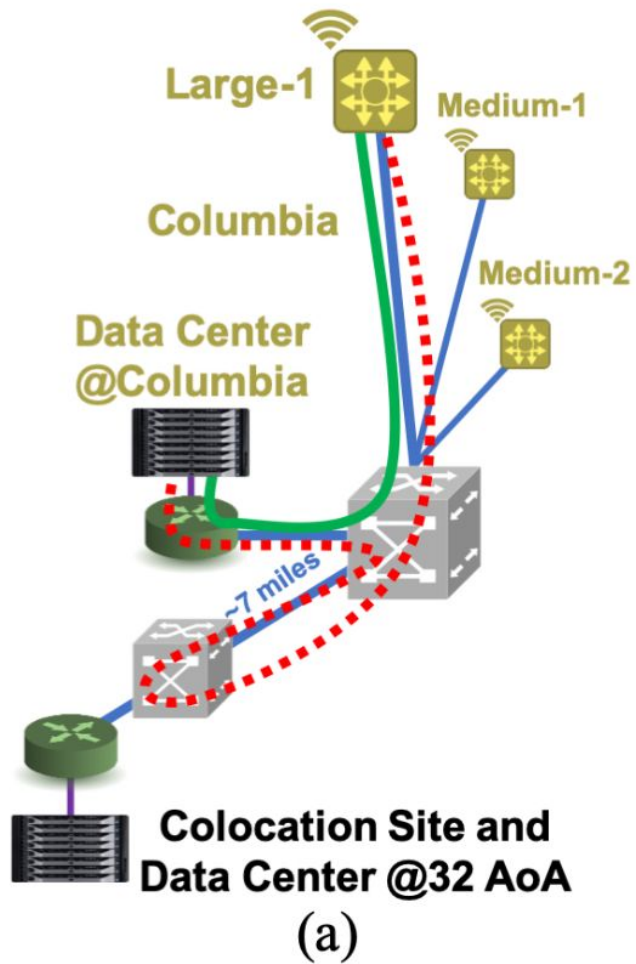
(c)

Figure 1. (a) The implemented Gen-2 RF canceller consisting of a wideband frequency-domain equalization (FDE) path and a narrowband path, (b) the measured TX/RX isolation of the Gen-2 RF canceller without turning on the canceller, and with the narrowband (Gen-1)/wideband (FDE) path on, when the circulator antenna port is terminated by 50 ohm (left) and connected to an antenna (right), and (c) the implemented open-access Gen-2 canceller box and wideband FD radios integrated in the COSMOS sandbox.



Experiment 2: Optical Wireless x-Haul Networking

- Wideband FD radio serves as a base station at the large node
- FD BS sends IQ data through core optical switching at Columbia data center to colocation side
- Sent back for digital screening
- Video multicast application operates on x-haul network on different optical wavelength
- Access points receive videos through on-demand optical switching managed by an SDN controller
- Evaluate ability of multicast application to switch between local and remote cloud servers
 - Depends on the amount of users connected



Experiment 3: Low Latency Edge Cloud

- Implication: computer intense resources like AR or VR will use on edge cloud to meet computing requirements
- Setup:
 - Configured in COSMOS optical network

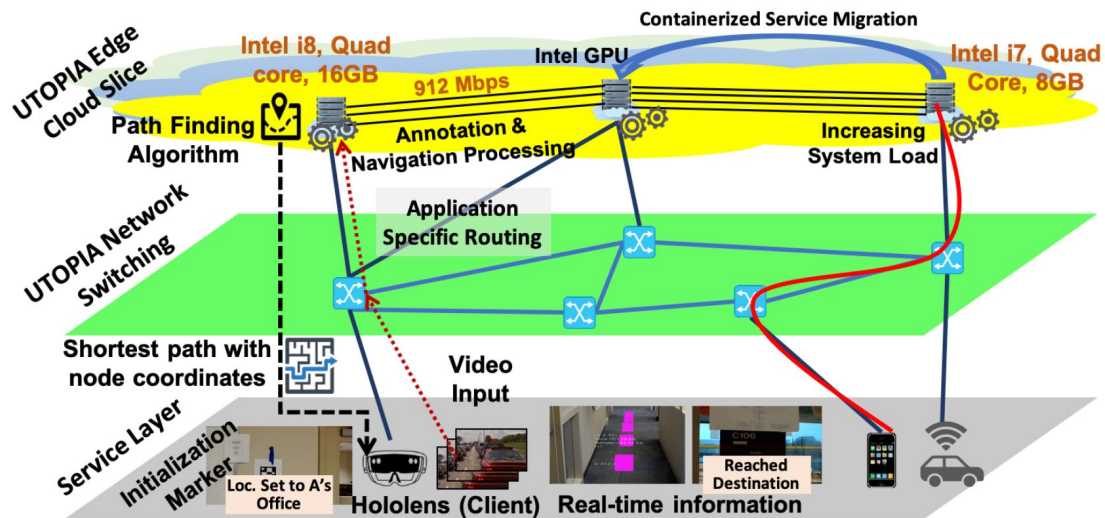


Figure 1. Setup of the low-latency AR edge cloud experiment, where the mesh topology can be dynamically configured by COSMOS' optical network.

Table 2: Comparison of the AR application end-to-end latency.

Normalized Average System Load	Same Edge Cloud	Edge Cloud w/ CSM	Edge Cloud w/ CSM and ASR
0.1	21.6 ms	21.6 ms	21.2 ms
0.4	89.7 ms	75.7 ms	58.5 ms
0.7	148.5 ms	110.0 ms	92.1 ms

Experiment 4: Smart City Intersections

- Autonomous driving a densely populated area can be tricky
 - Use COSMOS and AI to aid in this process
 - Send vehicle sensing data to a cloud where it is shared with other vehicles
- Testbed:
 - Custom deep learning algorithm to track all moving objects at an intersection
 - Send images and videos to monitoring applications to help keep city intersections safe
- First step towards new technologies that can either help control traffic flow or be used for autonomous vehicles later

